

Standing on Giants' Shoulders

Transfer learning — borrow a giant's brain, teach it your trick

Day 07 · Week 2

Unlocks  RPS Arena

1 Mission briefing

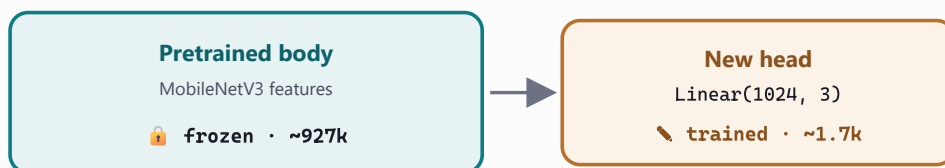
Training the Photo Detective from scratch was real work — and it still tops out near 70%. What if you could borrow a network that already spent weeks studying millions of images, and teach it just *one* new trick? That is **transfer learning**. Today you stand on a giant's shoulders: freeze a pretrained MobileNetV3, transplant a fresh head, and teach it rock-paper-scissors with photos you collect yourself. *Previously:* you unlocked  Photo Detective, a CNN that classifies real photos.

- ☐ Understand **transfer learning** (an expert chef, one new dish)
- ☐ Freeze the body, transplant **head** (~927k vs ~1.7k)
- ☐ Multiply your data **augmentation** (flip / rotate)
- ☐ Collect your own photos by webcam or phone
- ☐ Prove the giant matters: **pretrained vs random**
- ☐ Wire the model into the RPS referee

2 Field guide the concepts

Transfer learning. An expert chef learning one new dish does not relearn how to hold a knife. A network pretrained on a million photos already knows edges, textures, and shapes — knowledge that transfers to *your* problem for free.

Freeze the body, transplant the head. Keep the giant's feature extractor (its "body") frozen, and replace only the final classifier (its "head") with a fresh 3-class layer. That is ~927,000 frozen weights and only ~1,700 you actually train — fast, and it needs very little data.



Freeze the giant's body and train only a fresh 3-class head — roughly 1,700 weights instead of a million.

Free data, real proof. **Augmentation** flips and rotates each photo into several free training examples. And to *prove* the giant matters, train the same head twice — once on the pretrained body, once on a random one. The gap between them is the transferred knowledge, made visible.

 **Matrix Watch** — that frozen giant is millions of stored weights, and every forward pass through it is a stack of matrix multiplies. You just retrain the last one — the rest was precomputed for you.

3 Lab missions check each off in the notebook

1 Finish the validation pipeline ☐

Assemble the eval transform: `Resize(256) → CenterCrop(224) → ImageNet Normalize`.

Expect: tensors shaped (3, 224, 224) in the giant's color world.

2 Freeze the giant's brain ☐

Set `requires_grad = False` across the feature body.

Expect: trainable parameters collapse from ~2.5M to about 1.7k.

3 Perform the transplant ☐

Swap `classifier[3]` for `nn.Linear(1024, 3)`.

Expect: a model that now outputs exactly 3 scores.

4 Point the optimizer at the new head ☐

Give `Adam` only the parameters that still require grad.

Expect: training touch the head and leave the body untouched.

5 Build a giant with no experience ☐

Make the same architecture from random init and train the head.

Expect: the pretrained model crush the random one — proof of transfer.

6 Teach the model to referee ☐

Wire predictions into the rock-paper-scissors judge.

Expect: it call your move and pick a winner (try  cheat mode).

4 Cheat sheet transfer learning in PyTorch

```
mobilenet_v3_small(weights="DEFAULT")
net.features.requires_grad_(False)
model.classifier[3] = nn.Linear(1024, 3)
datasets.ImageFolder(root, transform=tf)
Resize(256) · CenterCrop(224)
Normalize(IMAGENET_MEAN, IMAGENET_STD)
RandomHorizontalFlip()
Adam(head.parameters(), lr=1e-3)
torch.jit.trace(model, ex).save(path)
```

Download a pretrained giant.

Freeze the whole body in one call.

Transplant a fresh 3-class head.

Read images; labels come from folder names.

Match the giant's expected input size.

Use the exact colors the giant learned on.

Free extra training data from a mirror.

Train only the unfrozen head.

Freeze the fine-tuned model to a file.

5 Unlock your Studio module

RPS Arena

Fine-tune, trace, and save your model. The unlock cell must write exactly:

```
ai_studio/models/rps_arena.pt
```

```
ai_studio/models/rps_labels.json
```

Then play rock-paper-scissors against your own model — and flip on  cheat mode:

```
$ python ai_studio/app.py
```

6 Mini-glossary

transfer learning

Reusing a model trained on one task to jump-start another.

fine-tuning

Lightly retraining part of a pretrained model on your data.

classifier head

The final layer(s) that turn features into class scores.

ImageNet

The million-image dataset most giants are pretrained on.

pretrained weights

The knowledge a network already learned on a huge dataset.

frozen layer

A layer whose weights are held fixed during training.

data augmentation

Making new training examples by transforming existing ones.

7 Show & tell

1. How many photos did you train on, and how good is your Arena? Does cheat mode feel fair?
2. Pretrained vs random init — how big was the gap? What does that prove about the giant?
3. Where does your model get confused (lighting, angle, background)? How would more data help?